

Assignment 1: Vectors and Vector Operations

Assigned: Jan 20, 2026 Tuesday

Due: Jan 30, 2026 Friday at 5:00pm

Rules (please read!!)

- **English:** Answer the questions in English. Otherwise, you'll lose half of the points.
- **Electronic submission:** Turn in solutions electronically via Blackboard. Be sure to submit your homework as a single file.
- **Late submission:** Submissions within 24 hours after the deadline: 20% penalty. Submissions after 24 hours of the deadline: 0 points awarded.
- **Academic Integrity Reminder:** Please ensure that you complete this assignment independently. If your solution appears to be identical or highly similar to others or generated using AI, you may be required to meet with the TA to verify your understanding.

Notations

- Scalars (real numbers) are denoted by lowercase letters such as $a, b, c \in \mathbb{R}$.
- Vectors are denoted by **bold** lowercase letters, such as $\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{a}, \mathbf{b}, \mathbf{c}$.
- Matrices (with real entries) are denoted by capital letters such as A , with entries $A_{ij} \in \mathbb{R}$.
- The vector $\mathbf{e}_i$ denotes the vector whose i -th entry is 1 and all other entries are 0. For example,

$$\mathbf{e}_1 = (1, 0, 0, \dots, 0), \quad \mathbf{e}_2 = (0, 1, 0, \dots, 0), \quad \mathbf{e}_3 = (0, 0, 1, \dots, 0).$$

- The vector of all ones of length n is denoted by $\mathbf{1}_n$.
- The vector of all zeros of length n is denoted by $\mathbf{0}_n$.
- We use $(a_1, a_2, \dots)$ to denote column vector and $[a_1, a_2, \dots]$ for row vector.

Total: 140 points, 8 problems.

1. Let $\langle \mathbf{v}, \mathbf{w} \rangle$ be the dot product of $\mathbf{v}, \mathbf{w}$. Verify that the following properties hold using the definition of dot products (20pts):

(a) Linearity:

$$\langle a\mathbf{u} + b\mathbf{v}, \mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{w} \rangle + b\langle \mathbf{v}, \mathbf{w} \rangle.$$

(5pts)

(b) Symmetry:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle.$$

(5pts)

(c) Positive definiteness: For any $\mathbf{u} \in V$,

$$\langle \mathbf{u}, \mathbf{u} \rangle \geq 0; \quad \text{and} \quad \langle \mathbf{u}, \mathbf{u} \rangle = 0 \iff \mathbf{u} = \mathbf{0}.$$

(5+5pts)

2. Verify that the vector norm defined in the lecture is indeed a norm, i.e. it satisfies the following (25pts):

(a) Non-negativity and non-degeneracy:

$$\|\mathbf{v}\| \geq 0; \quad \text{and} \quad \|\mathbf{v}\| = 0 \iff \mathbf{v} = \mathbf{0}.$$

(5+5pts)

(b) Absolute homogeneity:

$$\|c\mathbf{v}\| = |c| \|\mathbf{v}\|.$$

(5pts)

(c) Triangle inequality:

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Hint: Apply the Cauchy-Schwartz inequality.

(10pts)

3. *Reverse triangle inequality.* Suppose $\mathbf{a}$ and $\mathbf{b}$ are vectors of the same size. Show that

$$\|\mathbf{a} + \mathbf{b}\| \geq \|\mathbf{a}\| - \|\mathbf{b}\|.$$

(5pts)

Hint: Apply the triangle inequality to $(\mathbf{a} + \mathbf{b})$ and $(-\mathbf{b})$.

4. *Symptoms vector.* (10pts):

A vector $\mathbf{s}$ records whether each of 30 different symptoms is present in a medical patient. The i -th entry satisfies

$$s_i = 1 \quad \text{if the patient has symptom } i, \quad s_i = 0 \quad \text{if the patient does not.}$$

(a) Express the total number of symptoms the patient has using vector notation. (5pts)

(b) Express, using vector notation, the statement that the patient exhibits exactly 6 out of the first 15 symptoms. (5pts)

Remark: For example:

- The patient exhibits the 1st, 2nd, 5th, 8th, 10th, 14th, and 26th symptoms, but no others.
- As another example, the patient exhibits the 2nd, 6th, 7th, 8th, 10th, 11th, 18th, and 27th symptoms, but no others.

These examples are only for illustration; your answers should be written in general vector form.

Hint:

- (a) Define a suitable vector and use an inner product.
- (b) Use an equation involving the entries of $\mathbf{s}$.

5. *Interpretation of angles.* (25pts):

Denote $\angle(\mathbf{u}, \mathbf{v})$ as the angle between any two vectors $\mathbf{u}$ and $\mathbf{v}$. Suppose there are three types of fruit in the supermarket: orange, apple and banana. Every week, Alice buys 3 kilos of oranges, 2 kilos of apples and 4 kilos of bananas; Bob buys 6 kilos of oranges, 1 kilo of apples and 2 kilos of bananas; Charlie buys 2 kilos of oranges, 3 kilos of apples and 1 kilo of bananas.

The three vectors corresponding to their choices are

$$\mathbf{a} = (3, 2, 4), \quad \mathbf{b} = (6, 1, 2), \quad \mathbf{c} = (2, 3, 1).$$

- (i) Compute the two angles $\theta_1 = \angle(\mathbf{a}, \mathbf{b})$ and $\theta_2 = \angle(\mathbf{a}, \mathbf{c})$. Make sure to convert the angles to degree measurements (for example, write 90° instead of $\pi/2$). (15pts)

Hint: You can use the website Wolfram Alpha or Symbolab to compute the arccos function; see <https://www.wolframalpha.com/input?i=arccos+0>.

- (ii) Which one is larger? Provide an interpretation of this comparison. (5pts for interpretation)

Hint: The keyword you can use is “preference” or “taste”. Use your own words to describe the interpretation. The answer can be a few sentences.

- (iii) In general, when evaluating an object (e.g. a university, a restaurant, a cellphone), everyone has their own “weight vector”. Provide an interpretation of the angle between the weight vectors of two people. (5pts)

6. *Auto-regressive model.* (15pts):

Suppose that $z_1, z_2, \dots$ is a time series, where the number z_t gives the value in period (or time) t . For example, z_t could be the gross sales at a particular store on day t .

An *auto-regressive (AR) model* is used to predict z_{t+1} from the previous M values,

$$z_t, z_{t-1}, \dots, z_{t-M+1},$$

according to the formula

$$\hat{z}_{t+1} = (z_t, z_{t-1}, \dots, z_{t-M+1})^T \boldsymbol{\beta}, \quad t = M, M+1, \dots$$

where $\hat{z}_{t+1}$ denotes the AR model’s prediction of z_{t+1} , M is the memory length of the model, and $\boldsymbol{\beta}$ is a vector of coefficients.

For this problem, assume that the time period is daily and that $M = 10$. Thus, the AR model predicts tomorrow’s value using the values from the last 10 days.

For each of the following cases, give a short interpretation or description of the AR model *in plain English*, without referring to mathematical concepts such as vectors, inner products, or coordinates. You may use words like “today,” “yesterday,” or “ k days ago.”

(a) $\beta \approx \mathbf{e}_{10}$. (5pts)

(b) $\beta \approx 0.9 \mathbf{e}_1 + 0.1 \mathbf{e}_2$. (5pts)

(c) $\beta \approx 0.1 \mathbf{1}_{10}$. (5pts)

7. *Interpolation of rational functions.* A rational function of degree three has the form

$$f(t) = \frac{c_1 + c_2 t + c_3 t^2 + c_4 t^3}{1 + d_1 t + d_2 t^2 + d_3 t^3}$$

where $c_1, c_2, c_3, c_4, d_1, d_2, d_3$ are coefficients. (“Rational” refers to the fact that f is a ratio of polynomials. Another name for f is bi-cubic.) Consider the interpolation conditions

$$f(t_i) = y_i, \quad i = 1, \dots, K$$

where t_i and y_i are given numbers. Express the interpolation conditions as a set of linear equations in the vector of coefficients

$$\boldsymbol{\theta} = (c_1, c_2, c_3, c_4, d_1, d_2, d_3),$$

as $A\boldsymbol{\theta} = \mathbf{b}$. Give A and $\mathbf{b}$, and their dimensions. (20pts)

8. *Solve the system of linear equations.* Solve the following linear systems using Gauss-Jordan elimination. (20pts, 10 for each system)

Remark: You shall represent the systems with augmented matrices first. If you solved the system, but did not use augmented matrices, you would only get 1/2 of the points.

$$\begin{cases} 2x_1 + 5x_2 + 3x_3 = 10 \\ 2x_1 + x_2 + 3x_3 = 6 \\ x_1 + 2x_2 + x_3 = 4 \end{cases} \quad (1)$$

$$\begin{cases} 5x_1 + 4x_2 + x_3 = 17 \\ x_1 + x_2 + 2x_3 = 9 \\ 3x_1 + 2x_2 + x_3 = 11 \end{cases} \quad (2)$$